HARA

HazID	Contents	Severity	Exposure	Controllability	AgPLr	Safety goal
PTO-HZ-000	**Hazard Analysis and Risk Assessment — Rear PTO** HARA per ISO 25119-2:2018, Clause 6, for the faulted UoO (malfunctioning behaviour of the PTO control). Participants: E. Vidal (safety), H. Larsen (systems), C. Reid (field/operator view), P. Ferraro (hardware), G. Hoffmann (assessor, observer). Harm to persons only; property-only outcomes are recorded as QM. Method: system states and transitions per the state flowchart in the Technical Safety Concept; for every malfunction the relevant operating situations were rated. Where frequency and duration methods give different exposure classes, the higher class is used (6.3.5).	S0	E0	C0	QM	—
—	**Risk graph** **AgPLr determination (ISO 25119-2, Figure 1)** S severity · E exposure · C controllability — C0 is always QM a…e = required agricultural performance level (AgPLr); QM = quality measures *Figure 1 — AgPLr determination (ISO 25119-2 Fig. 1); rated cells of this HARA are ringed*	S0	E0	C0	QM	—
PTO-HZ-01	Unintended PTO engagement while a person is at the implement or driveline (coupling the shaft, fitting the guard, adjusting the machine; engine running, PTO commanded OFF). Harm scenario: clothing or limb caught by the engaging shaft/implement intake — entanglement injuries.	S3	E3	C3	AgPL d	PTO-SG-01
PTO-HZ-02	Unintended engagement with a bystander next to a stationary implement (slurry pump, auger) — person feeding or clearing the machine while PTO is OFF.	S3	E2	C3	AgPL c	PTO-SG-01
PTO-HZ-03	PTO fails to disengage on operator stop command. Typical scenario: baler pickup blocks, operator commands PTO OFF and dismounts to clear it while the shaft is in fact still driven.	S3	E3	C2	AgPL c	PTO-SG-02
PTO-HZ-04	PTO does not shut off when the operator leaves the seat during mobile work (operator-presence function fails), operator walks to the still-driven implement.	S3	E2	C3	AgPL c	PTO-SG-03
PTO-HZ-05	Automatic re-engagement at key-on / engine restart with the cab switch still latched from the previous session; person at the implement during the morning start.	S3	E2	C3	AgPL c	PTO-SG-03
PTO-HZ-06	Fender jog latches ON or fails to stop on button release — the operator standing at the fender is within arm's reach of the turning shaft end.	S3	E2	C2	AgPL b	PTO-SG-04
PTO-HZ-07	Shaft overspeed: 1000-mode speed delivered while a 540-only implement is attached (wrong internal mode or wrong ratio supervision). Implement overspeed → break-up, ejected parts hitting a bystander.	S3	E1	C3	AgPL b	PTO-SG-05
PTO-HZ-08	Engagement without torque ramp (full pressure step): driveline shock can shear shaft bolts / yoke, fragments near the hitch area.	S2	E2	C3	AgPL b	PTO-SG-05
PTO-HZ-09	PTO state indication wrong (cluster lamp shows OFF while shaft is driven). Contributes to PTO-HZ-03-type scenarios; standalone harm requires additional failures.	S2	E1	C2	QM	—
PTO-HZ-10	Unintended engagement during road transport with no implement attached (stub shaft turns under its guard).	S1	E2	C2	QM	—
—	**Result** Highest requirement: AgPL d (PTO-HZ-01). Safety goals PTO-SG-01 … PTO-SG-05 are derived in the Functional Safety Concept; every hazardous situation with AgPLr > QM is covered by exactly one safety goal. This HARA is reworked on any change of the UoO scope (impact analysis per ISO 25119-4 Clause 11).	S0	E0	C0	QM	→ FSC